

Persistent Memory with Fixed Geometry Scoring

An OpenJev Reacher development study

OpenJev project · September 19, 2026 (UTC)

Development report; not an ICLR submission or an independently replicated result.

Abstract. We evaluate whether persistent learned state remains useful when Reacher controllers share a fixed geometry-based scoring rule. Twelve inherited models, comprising three paired fits of four learned families, are evaluated on 64 fresh paired cases in each of three sensing conditions. No models are fitted in this study. Persistent GRU reduces mean native episode cost by 31.86% under six-step sensing gaps and 25.99% under ten-step gaps relative to a GRU that resets its hidden state and retains the last observed angles. All 25 frozen continuation checks pass. Supplied-physics controllers still outperform every learned fit. The result supports these trained persistent update policies over the declared reset controls in this task; it does not establish a novel architecture, biological mechanism, calibrated uncertainty or general robotics capability. A trained two-valid-observation/action-history control remains unresolved.

1 Comparison and task

The environment is Gymnasium Reacher-v5 with native reward weights 1 for distance and control cost, 50 actions per episode and $\Delta t = 0.02$ seconds. An issued command $a_t \in [-1, 1]^2$ receives independent hidden Gaussian actuator noise with standard deviation 0.05 and is clipped again before execution. Reported utility is native episode cost $C = -\sum_{t=0}^{49} r_t$, so lower is better. The learned controller sees

$$p_t = [\cos q_0, \cos q_1, \sin q_0, \sin q_1, g_x, g_y, \text{valid}, \text{age}],$$

with unavailable angle entries masked to zero. Native velocity, realized noise and fingertip displacement are excluded. The public target remains known. Conditions are full sensing and two gaps of length six or ten; gap onset phase varies by case. Full sensing is still partially observed because velocity is withheld.

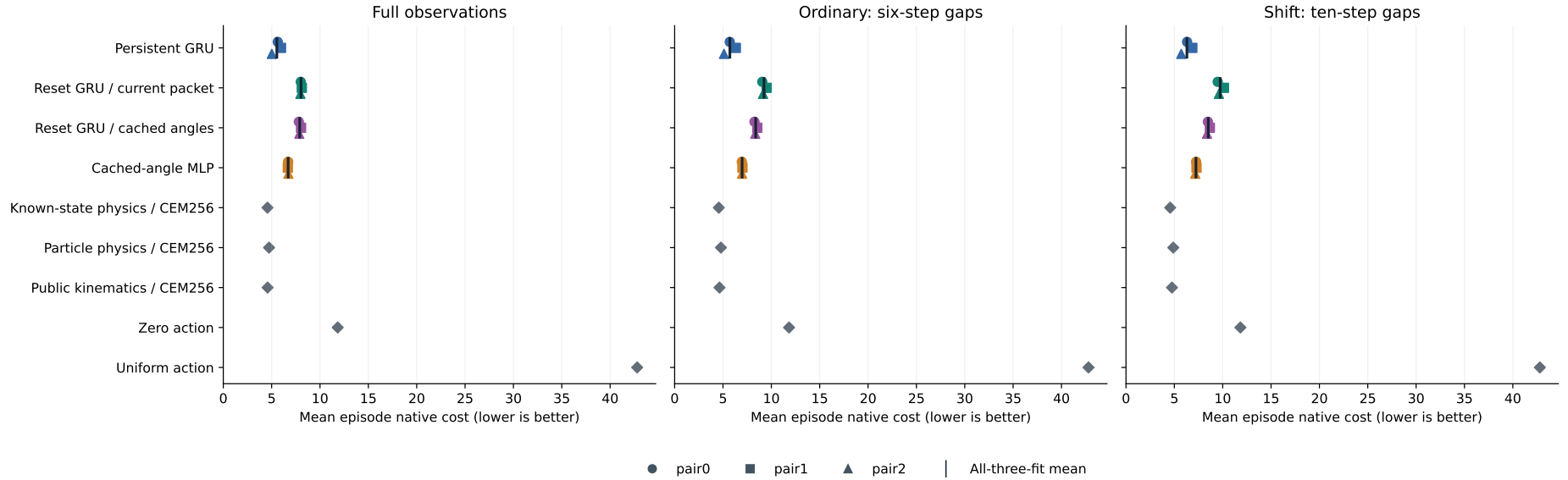
The GRU families share the same 36,805-parameter schema and paired original initialization, training corpus, order and update count, but their final weights were trained separately. Persistent GRU carries learned state across real observations and actions. Current-packet GRU resets and encodes the actual packet; cached-observation GRU resets and encodes the last measured angles with current goal, validity and age. Cached MLP has 36,599 parameters and the same public angle cache. All GRUs remain recurrent in imagined rollouts. The treatment includes the trained real-observation update policy and validity gating, rather than recurrence in isolation.

Table 1: Mean native episode cost. Learned families average all three saved fits; each reference is evaluated on the same 64 cases. No best-fit selection.

Controller	Full sensing	Six-step gaps	Ten-step gaps
Persistent GRU	5.537576	5.709233	6.296307
Current-packet GRU	8.034106	9.240609	9.740743
Cached-observation GRU	7.902708	8.378622	8.507414
Cached-observation MLP	6.705506	6.975331	7.234155
Known-state physics	4.568717	4.568717	4.568717
Particle physics	4.731065	4.786677	4.892595
Public-kinematic physics	4.580680	4.640030	4.755555
Zero action	11.830064	11.830064	11.830064
Uniform action	42.796284	42.796284	42.796284

2 All fits retained; persistent update policies improve cost

OpenJev: real-observation memory policies, fixed geometry scoring



All 36 learned points and 15 references. All planned rows use CEM256; floors use no planner. No best-fit selection.

Figure 1: All 36 learned fit/panel points and 15 reference points. Shapes identify the three saved fit pairs; ticks show their family mean. References retain distinct information privileges. No error bars over three fits or episode-by-episode dominance are implied.

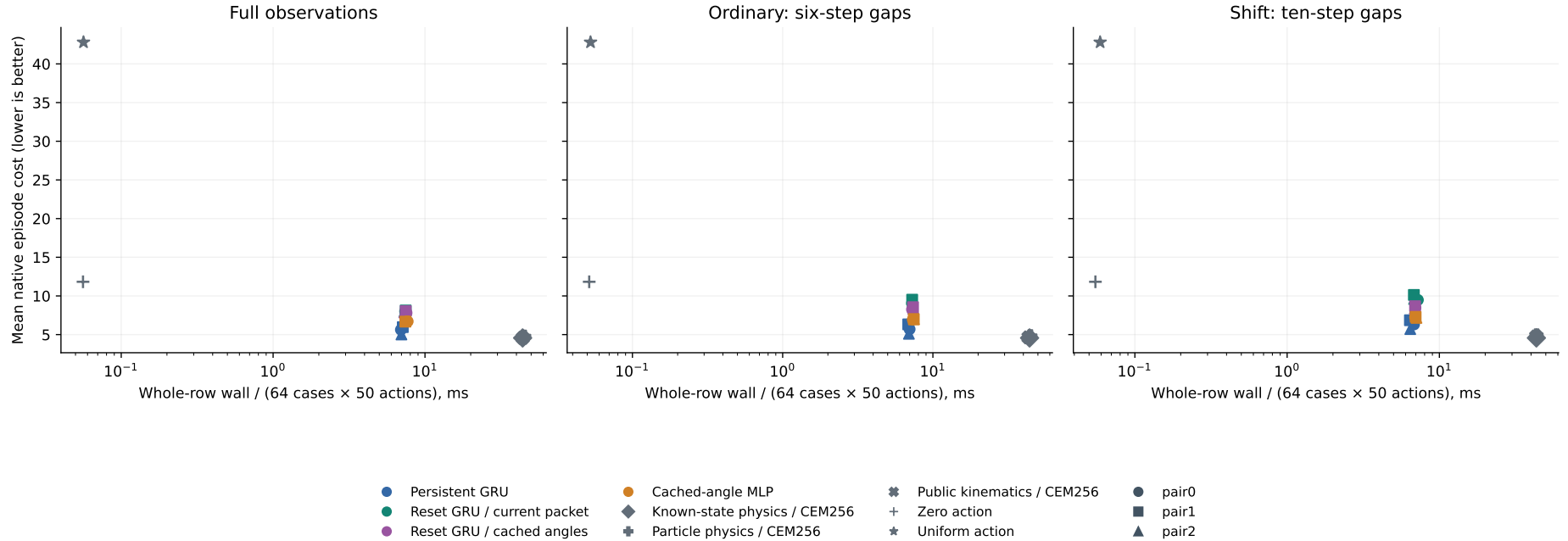
Against current-packet GRU, persistent GRU lowers mean cost by 38.22% and 35.36% on the two gap panels. Against cached-observation GRU, the corresponding reductions are 31.86% and 25.99%; all three paired fits improve in both panels. The full-sensing reductions are 31.07% and 29.93%, respectively, so the benefit is not confined to missing measurements. This experiment does not identify velocity inference as the mechanism.

For persistent minus cached GRU, the saved conditional paired-case 95% cost-difference intervals are $[-3.1291, -2.2102]$ for six-step gaps and $[-2.5642, -1.8672]$ for ten-step gaps. These intervals hold the three fits and training corpus fixed; they are neither training-seed population uncertainty nor multiplicity-corrected confirmation. No new resampling was performed for this report.

Persistent GRU also improves over cached MLP by 18.15% and 12.96% on the gap panels, a secondary result that cannot rescue the primary rule. All three supplied-physics references are better than every learned fit in every panel. Persistent cost is 23.04% and 32.40% higher than public-kinematic physics on the gap panels. These physics references use supplied nominal dynamics; known-state physics additionally receives current true position and velocity. They are not optimal stochastic-control oracles.

3 Fixed scoring and candidate budgets, unequal total computation

Utility versus measured control cost: every fit and reference



Shared host; setup, native steps and traces included. Amortized throughput, not isolated latency, matched total compute or a selected frontier.

Figure 2: All learned fits and references, with no selected frontier. The horizontal coordinate is complete recorded row wall time divided by 64×50 , on a log scale. This is amortized shared-host throughput, not isolated action latency or a cross-machine speed result.

Every nonfloor controller uses geometry scoring and CEM256 with horizon 12, three-action blocks, a paid mean candidate, global-best retention and terminal truncation. Geometry projects predicted trigonometric features with atan2 , uses 0.10/0.11 m link offsets, and subtracts expected clipped actuator cost once. Predicted per-step rewards are clipped to $[-2.5, 0]$ for planning. Original learned reward heads still execute and are charged. Native episode cost is reported without replacing it by this planning score.

Reset states, hidden disturbances and planner innovations are paired across rows; score-adaptive CEM proposals can subsequently differ. Equal candidate budgets do not match information or total compute. Persistent rows take 20.49 to 22.85 s; other learned rows, 21.76 to 24.62 s; supplied-physics rows, 138.31 to 143.96 s. These times include controller setup, native stepping and trace storage. Global source/checkpoint validation and copying are separate execution costs. The complete execution takes 2,109.20 s within its frozen 5,700 s cap; audit takes 1,384.44 s within its separate 3,300 s cap.

4 Frozen decision rule and verification boundary

The study retains all twelve inherited fits, all 51 control rows and all 64 cases per row. There are zero new fits, optimizer updates or Astra calls. The frozen rule requires persistent GRU to improve both reset-GRU family means by at least 3% on both gap panels, be no worse in any paired fit, and degrade by no more than 2% on full sensing. Persistent fits on both gap panels and known-state physics on ordinary gaps must each beat zero action by at least 10%. All **25/25 checks pass** without threshold changes.

Table 2: Audited paid work and replay scopes. Selected one-step predictions are additional to the candidate trajectories; nested work is not additive wall time.

Scope	Count
Learned candidate sequences	29,491,200
Learned imagined candidate transitions	314,966,016
Learned selected one-step advances	115,200
Physics candidate sequences	7,372,800
Physics nominal candidate transitions replayed	78,741,504
Physics selected advances replayed	28,800
Executed native control transitions replayed	163,200
Public-observer transitions checked	310,464

The saved-output auditor authenticates the 90 frozen sources, checkpoint lineage, all 9,505 execution members, public-history boundaries, chosen actions and search arithmetic. Native-control, nominal-planning and public-observer replays each have maximum absolute discrepancy zero. The audit makes **zero learned-model calls**: saved neural predictions and hidden states remain source-bound evidence, not independently recomputed neural outputs. Equal before/after weights alone do not prove the absence of intermediate updates. Publication uses saved artifacts only and adds no scientific calls.

5 Interpretation and next control

This is fresh-case evaluation of previously trained models in a repeatedly studied simulator, not an independent training replication. The common geometry score approximates native RK4 cached-body reward; distance at projected mean angles is not expected distance. Neither the physics advantage nor the learned gap quantifies optimal control. Training data and paired initialization reduce some confounds, but differing trained update rules, validity gates and final weights prevent attributing the result to recurrence alone.

The next conventional control is a parameter-identical GRU trained to rebuild state from its last two valid observations and all intervening issued actions. That trained comparison is unresolved. Component tests do not supply an effectiveness result. A separate protocol, complete runner/audit, rehearsal and capacity measurement are required before fitting or evaluation. A second environment and fresh training replication remain separate evidence needs. No connectome or biological advantage, Bayesian inference, calibrated uncertainty, architectural novelty or general robotics performance is established.

Evidence and artifact identity

[Study report](#); [all fit-level results and conditional intervals](#); [frozen protocol](#); [audit receipt](#). These links locate the project artifacts; the following SHA256 identities bind the exact local inputs used here. The paper is a summary, not a complete raw-trace archive.

Plan	23c93e4adfb45cf224383ffa31d7323df2405807eb3f1c4c75bb27378bcd9ee
Audit	cb64ded7bbcc4ad6e342f9ac3b431e085e754f77075bb2e4852d6b82d29a54c6
Summary	921c7f12f73169dd788f8e8204ee5377949e3a5c8b7cd88023887a4da6e5753d